Breast Cancer Research and Treatment (2020) 180:1–19
https://doi.org/10.1007/s10549-019-05516-1

REVIEW

Uterine cancer in breast cancer survivors: a systematic review

Akemi T. Wijayabahu[1] · Kathleen M. Egan[2] · Lusine Yaghjyan[1]

Received: 29 August 2019 / Accepted: 21 December 2019 / Published online: 2 January 2020

Abstract

Purpose Epidemiological evidence on the risk factors for uterine/endometrial cancer in breast cancer (BCa) survivors is limited and inconsistent. Therefore, we critically reviewed and summarized available evidence related to the risk factors for uterine/endometrial cancer in BCa survivors.

Methods We conducted a literature search through PubMed, Web of Science Core Collection/Cited Reference Search, as well as through manual searches of the bibliographies of the articles identified in electronic searches. We included in this review studies that were published up to November 30, 2018 that were accessible in full-text format and were published in English.

Results Of the 27 eligible studies, 96% had > 700 participants, 74% were prospective cohorts, 70% originated outside of the US, 44% reported as having pre-/postmenopausal women, and 26% reported having racially heterogeneous populations. Risk factors positively associated with uterine/endometrial cancer risk among BCa survivors included age at BCa diagnosis > 50 years, African American race, greater BMI/weight gain, and Tamoxifen treatment. For other lifestyle, reproductive and clinical factors, associations were either not significant (parity) or inconsistent (HRT use, menopausal status, smoking status) or had limited evidence (alcohol intake, family history of cancer, age at first birth, oral contraceptive use, age at menopause, comorbidities).

Conclusion We identified several methodological concerns and limitations across epidemiological studies on potential risk factors for uterine/endometrial cancer in BCa survivors, including lack of details on uterine/endometrial cancer case ascertainment, varying and imprecise definitions of important covariates, insufficient adjustment for potential confounders, and small numbers of uterine/endometrial cancer cases in the overall as well as stratified analyses. Based on the available evidence, older age and higher body weight measures appear to be a shared risk factor for uterine/endometrial cancer in the general population as well as in BCa survivors. In addition, there is suggestive evidence that African American BCa survivors have a higher risk of uterine/endometrial cancer as compared to their White counterparts. There is also evidence that Tamoxifen contributes to uterine/endometrial cancer in BCa survivors. Given limitations of existing studies, more thorough investigation of these associations is warranted to identify additional preventive strategies needed for BCa survivors to reduce uterine/endometrial cancer risk and improve overall survival.

Keywords Breast cancer survivors · Uterine cancer · Endometrial cancer · Systematic review

Abbreviations

ACS	American Cancer Society
BCa	Breast cancer
CI	Confidence interval
HR	Hazard ratios
OR	Odds ratio
RR	Relative risk
SEER	Surveillance, epidemiology, and end results
SIR	Standardized incidence ratio
US	United States

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s10549-019-05516-1) contains supplementary material, which is available to authorized users.

✉ Lusine Yaghjyan
lyaghjyan@ufl.edu

1 Department of Epidemiology, College of Public Health and Health Professions and College of Medicine, University of Florida, 2004 Mowry Rd, Gainesville, FL 32610, USA

2 Department of Cancer Epidemiology, H. Lee Moffitt Cancer Center & Research Institute, 12902 USF Magnolia Drive, Tampa, FL 33612, USA

Introduction

According to the American Cancer Society (ACS), breast cancer is the second most common cancer in women in the United States (US) [1]. Advances in breast cancer detection and treatment have resulted in improved survival with a 90% five-year relative overall survival after diagnosis [2]. Based on data from Surveillance, Epidemiology and End Results (SEER), more than 3.5 million women in the United States (US) were breast cancer survivors in 2016 [2]. As compared to the general population, breast cancer survivors are at an increased risk of developing any secondary cancer with an excess risk of 30% (excluding contralateral breast cancer) [3]. Uterine cancer, and especially endometrial cancer, is the most frequently observed secondary malignancy among breast cancer survivors across studies from various countries [3–6] and racial/ethnic groups [7]. Endometrial cancers (endometrial adenocarcinomas) comprise about 80–90% of all uterine malignancies and remaining 20–10% include adenosquamous carcinomas, papillary serous carcinomas, and uterine sarcomas [8]. A prospective study conducted by Ricceri et al. found an increase in endometrial cancer incidence among breast cancer survivors (Standardized Incidence Ratio [SIR] 2.18, 95% Confidence Interval [CI] 1.75–2.70) [3]. Another prospective study conducted by Li et al. reported a higher risk of uterine cancer among breast cancer survivors (SIR 1.17, 95% CI 1.13–1.21) [9]. Further, the overall 5-year survival in women diagnosed with endometrial cancer after breast cancer diagnosis ranged between 61 and 73% in contrast to 82% survival rate in women with endometrial cancer who did not have prior breast cancer [10].

Breast cancer survivors may be at an increased risk of developing uterine cancer, especially endometrial cancer due to shared risk factors including obesity [11–13], underlying genetics [14, 15], and hormonal factors [9, 16]. Moreover, hormonal treatment for breast cancer could affect risk of a subsequent hormone-dependent tumor [17, 18]. However, there is limited epidemiologic evidence on the risk factors for uterine cancer among breast cancer survivors [7, 9, 16, 19, 20]. Thus, there is an urgent need to review and summarize the current knowledge and identify research gaps with regards to the potential risk factors for uterine cancer in breast cancer survivors in order to identify any additional preventive strategies that might be needed in breast cancer survivors to reduce the risk of uterine cancer and improve overall survival. We critically reviewed epidemiologic evidence related to the risk factors for uterine cancer in breast cancer survivors and assessed strengths and limitations of existing studies to identify research gaps and suggest potential areas for future research.

Methods

Literature search

Relevant articles were identified using PubMed (US National Institutes of Health [NIH]) and Web of Science Core Collection/Cited Reference Search. We included in this review original research articles published before November 30, 2018 that investigated any risk factors for uterine cancer among female breast cancer survivors, were accessible in full-text format, and were published in English.

In this review, we included studies that investigated any risk factors for any type of uterine cancers as well as articles that focused on endometrial cancer only. Articles were first searched in PubMed using combinations of the term "Breast cancer survivor" with any of the following MeSH terms for uterine cancer: "Endometrial Cancer," "Endometrial carcinoma," "Endometrial neoplasm," "Uterine Cancer," "Uterine corpus," "Uterine carcinoma," "Uterine neoplasm" (see more detail in Supplemental Table 1). Once eligible articles were identified, a second search was conducted on Web of Science Core Collection/Cited Reference Search by including the titles of the selected manuscripts in order to capture eligible research articles that were cited in the original reports. Bibliography of the articles found through electronic searches were used to identify additional relevant references that were then hand-searched. We excluded articles that evaluated the risk of uterine cancer prognosis, articles reporting effects of herbal treatment after breast cancer diagnosis, and articles that reported risk estimates for gynecological cancers combined rather than specifically for uterine/endometrial cancer. After removing duplicates from the combined PubMed, Web of Science and bibliography search, one reviewer assessed eligibility of the articles which was next verified by the second reviewer.

Next, information on study features was abstracted by the first reviewer and subsequently confirmed by the second reviewer. From each article, we extracted information on study design, sample size, characteristics of the study population, outcome ascertainment methods, and potential risk factors investigated in relation to the risk of uterine cancers or endometrial cancer specifically. We discussed study findings for each of the investigated risk factors and identified potential biases. Finally, we compared results of this review with what is known about risk factors for primary uterine/endometrial cancer in women without cancer history and identified knowledge gaps in this field of study.

Statistical analysis

Heterogeneity of the studies with respect to the study populations and presentation of the risk estimates makes a direct comparison across the studies difficult. For these reasons, we were unable to conduct a formal meta-analysis across the studies. For the studies that reported estimates for the risk factor of interest only while stratifying by another factor, we extrapolated the relevant overall risk estimates by calculating the average estimate weighted by the number of participants in each stratum.

We summarized the associations of investigated risk factors with uterine/endometrial cancer risk among breast cancer survivors in Figs. 2, 3, and 4 for studies reporting Hazard ratios (HR), Relative risk (RR), or Odds ratio (OR) and Fig. 5 for studies reporting Standardized Incidence Ratios (SIR). We excluded studies by Raymond et al. [21] (selectively reported significant findings for specific levels of risk factors) and Nsouli-Maktabi et al. [7] (did not report risk estimates) from the figures. The figures were created with the online DistillerSR Forest Plots Generator tool [22].

Results

Characteristics of the studies

Our search yielded 160 manuscripts from which we identified 27 eligible studies (Fig. 1). Key characteristics of the studies are summarized in Table 1. The earliest study was published in 1998. Majority of published studies originated outside of the US ($n = 19$, 70%). Of the 8 studies that were published in the US, 5 utilized SEER data. Most of the studies were prospective cohorts ($n = 20$, 74%) with an average follow-up duration of 28 years (range 11–63 years); four studies utilized case–control design, two studies were retrospective cohorts, and one study was based on retrospective chart review. Twelve of the studies included both pre-/perimenopausal and postmenopausal women (44%). Of the studies with both pre- and postmenopausal women, 6 reported findings separately by menopausal status.

Out of the 15 studies with defined or known racial/ethnic composition, 7 had a racially/ethnically heterogeneous study population; 6 studies included Asian participants and 2 studies included predominantly white women. The majority of the studies had a large sample size (> 700 participants) with the average number of participants equal to 249,787 (range 2786–3,766,825 for cohort studies and 622–1880 for case–control studies). For each study, we present characteristics of the study population and details on other design features in Table 2. The majority of the studies utilized medical records to ascertain uterine/endometrial cancer cases among breast cancer survivors, while one study used health insurance records [23] and another used self-reported online questionnaire [19]. Fourteen out of twenty-seven studies (52%) explored risk factors for endometrial cancer alone and the rest of the studies (48%) investigated risk factors for uterine cancer combining tumor types.

Table 1 Selected characteristics of the 27 studies on the risk factors of uterine/endometrial cancer in breast cancer survivors

Study characteristics	*N* (%)	Sample size range	Total sample size across all studies
Study design			
Prospective cohort	20 (74)	2786–3,766,825	6,668,261
Retrospective cohort	2 (7)	21,527–74,280	95,807
Retrospective chart review	1 (4)	–	732
Case–control[a]	4 (15)	622–1880	4,699
Menopausal status[b]			
Premenopausal/perimenopausal	11 (41)	123–204,149	289,098
Postmenopausal	11 (41)	386–390,888	772,693
Postmenopausal/perimenopausal	1 (4)	–	1,019
Heterogeneity of the study population[c]			
Predominantly white	2 (7)	1038–10,953	11,991
Homogeneous, Asian	6 (22)	2786–74,280	254,065
Heterogeneous	7 (26)	732–525,527	1,295,237
Undefined	13 (48)	622–3,766,825	4,621,280

[a]Sample size reflects total number of cases and controls combined

[b]Numbers are based on the studies with reported numbers by menopausal status

[c]Studies which did not report the ethnic/racial distribution of the population were categorized as “undefined” except the studies that were conducted in countries that are predominantly homogeneous (Taiwan, Korea, Japan) and the study conducted by Mellemkjaer et al. [28] which was considered to have a heterogeneous study sample (inclusion of multiple countries with diverse racial/ethnic origins)

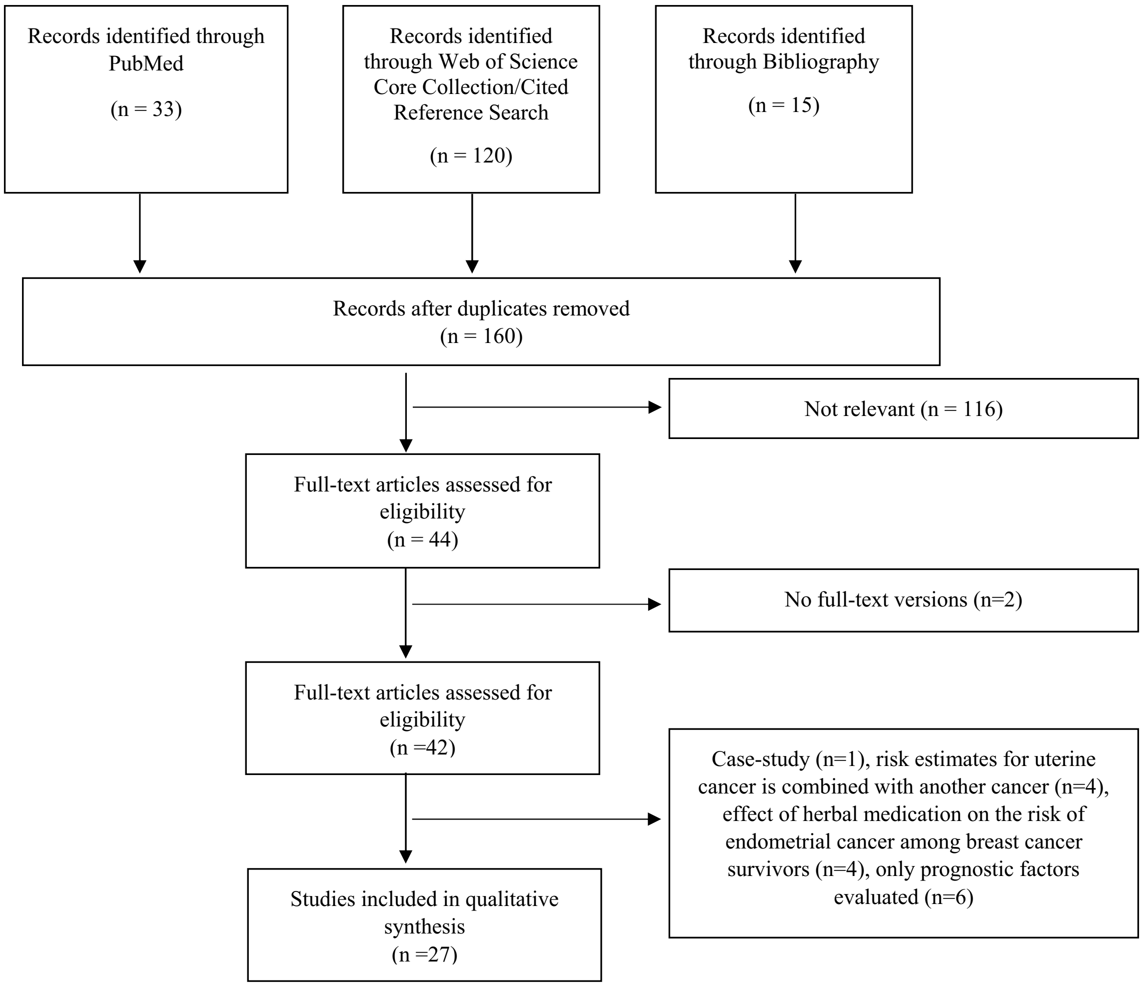

Fig. 1 Flow diagram of the literature search

Risk factors of uterine and endometrial cancer among breast cancer survivors

Most of the studies simultaneously evaluated more than one potential risk factor for uterine or endometrial cancer among breast cancer survivors, totaling 15 studies for age at breast cancer diagnosis, 13 studies for adjuvant hormone therapy after breast cancer diagnosis (with Tamoxifen evaluated in 9 of these studies), 7 studies for other therapy types (radiotherapy and chemotherapy), 6 studies for menopausal status, and 3 studies for race, smoking status, family history of cancer, and hormone replacement therapy (HRT) (Table 2). Information on other lifestyle or reproductive risk factors (adult weight gain, Body Mass Index [BMI], body weight, alcohol intake, age at menarche, parity, age at first birth, oral contraceptive use, age at menopause), breast tumor characteristics (molecular subtype, histology, grade, stage etc.), and comorbidities (diabetes, hypertension, thyroid disease etc.) were investigated in two or fewer studies. We summarized the associations of investigated risk factors with uterine/endometrial cancer risk among breast cancer survivors in Figs. 2, 3, 4 (for HR/RR/OR) and Fig. 5 (for SIR).

Eleven out of thirteen studies that reported the risk of uterine or endometrial cancer by the age at breast cancer diagnosis (Figs. 2 and 5) found a statistically significant elevated risk with increasing age, especially after age of 50 years [5, 6, 9, 24–31]. Two studies that reported the associations between race and uterine cancer among breast cancer survivors (Figs. 2 and 5) showed an increased risk of uterine cancer among African American as compared to white women [9, 20], while one study reported a significantly higher cumulative incidence of endometrial cancer among white breast cancer survivors as compared to African American survivors [7]. Two out of three studies that reported the associations of body weight (BMI at breast cancer diagnosis) [12, 19], adult weight gain since breast cancer diagnosis [12], body weight at breast cancer diagnosis [32] found an increased risk of uterine/endometrial cancer among obese ($\geq$ 28.9 vs. < 22.5 kg/m^2) [12] breast cancer survivors and those with a heavier weight (> 70 vs. < 60 kg) [32] as

Table 2 Summary of the 27 studies on the risk factors for uterine/endometrial cancer in breast cancer survivors

Author (year)	Country	Study design (N)	Menopausal status at BCa diagnosis (N pre- or peri/postmenopausal)	Breast cancer ascertainment (type of BCa)	Uterine cancer ascertainment	Number of Uterine / endometrial cancer in the analysis	Risk factors explored in the study
Li et al. [9]	USA	Prospective, 666,262	NA	Medical records (NA)	Medical records	Uterine cancer ($n=33033$)	Age at BCa diagnosis, race, menopausal status, breast tumor characteristics: receptor status (ER/PR status)
Chen et al. [24]	Taiwan	Prospective, 86,827	NA	Medical records (NA)	Medical records	Uterine cancer ($n=238$)	Age at BCa diagnosis
Corso et al. [6]	Italy	Retrospective cohort, 21,527	10,377/11,150[a]	Medical records (Invasive BCa)	Medical records	Uterine and Endometrial cancer ($n=98$)	Age at BCa diagnosis, family history of breast cancer/ovarian cancer (first degree family history), menopausal status at BCa diagnosis, breast tumor characteristics: receptor status (ER/PR status, HER-2 status, luminal), histology, grade, stage, lymph node involvement (number of positive nodes), perivascular invasion, adjuvant hormone therapy (not specified), chemotherapy, radiotherapy
Jung et al. [4]	Korea	Prospective, 3344	NA	Medical records (NA)	Medical records	Endometrial cancer ($n=6$)	Adjuvant hormone therapy (not specified), radiotherapy
Chlebowski et al. [38]	USA	Prospective, 17,064	–/17,064[a]	Medical records (Invasive BCa)	Medical records	Endometrial cancer ($n=148$)	Adjuvant hormone therapy (Tamoxifen and Aromatase Inhibitors)
Liu et al. [20]	USA	Prospective, 289,933	NA	Medical records (Invasive BCa)	Medical records	Endometrial cancer ($n=2044$)	Age at BCa diagnosis, race
Ricceri et al. [3]	Denmark, France, Germany, Greece, Italy, Netherlands, Norway, Spain, Sweden, UK	Prospective, 11,045	4,963/5,736a	Medical records and/or health insurance records (NA)	Medical records and/or health insurance records	Uterine cancer ($n=39$)	Breast tumor characteristics: stage

Table 2 (continued)

Author (year)	Country	Study design (N)	Menopausal status at BCa diagnosis (N pre- or peri/postmenopausal)	Breast cancer ascertainment (type of BCa)	Uterine cancer ascertainment	Number of Uterine / endometrial cancer in the analysis	Risk factors explored in the study
Torres et al. [19]	USA	Case /Control study, 173 BCA survivors with ECA/ 865 BCA survivors without)	Cases: 544/307[a] Controls: 82/79[a]	Online questionnaire (In situ and invasive BCa)	Online questionnaire	Uterine cancer ($n = 173$)	Age at uterine cancer diagnosis, BMI, smoking history, family history of colon/uterine cancer, HRT use, menopausal status, comorbidities (diabetes, hypertension, thyroid disease, gallbladder disease), Adjuvant hormone therapy (Tamoxifen use and duration of use)
Chen et al. [23]	Taiwan	Retrospective cohort, 74,280	NA	Health insurance records (NA)	Health insurance records	Endometrial cancer (222)	Comorbidities (diabetes, hypertension), adjuvant hormone therapy (Tamoxifen use)
Molina-Montes et al. [5]	Spain	Prospective, 5897	1727/4170	Medical records (Invasive BCa)	Medical records	Endometrial cancer ($n = 44$)	Age at BCa diagnosis
Kamigaki et al. [29]	Japan	Prospective, 33,045	NA	Medical records (Invasive BCa)	Medical records	Uterine cancer ($n = 53$)	Age at BCa diagnosis, breast tumor characteristics: lymph node involvement (lymph node metastasis), adjuvant hormone therapy (Not specified), chemotherapy, radiotherapy
Mellekiaer et al. 2011	Denmark, Finland, Norway	Prospective, 304,703	NA	Medical records (Invasive BCa)	Medical records	Endometrial cancer ($n = 2,192$)	Age at BCa diagnosis
Nsouli-Maktabi et al. [7]	USA	Prospective, 450,936	60,048/390,888[a]	Medical records (In situ and invasive BCa)	Medical records	Endometrial cancer ($n = 3,827$)	Age at BCa diagnosis, race
Bland et al. [16]	USA	Retrospective chart review, 732	NA	Medical records (NA)	Medical records	Endometrial cancer ($n = 59$)	Adjuvant hormone therapy (duration of Tamoxifen use)

Table 2 (continued)

Author (year)	Country	Study design (N)	Menopausal status at BCa diagnosis (N pre- or peri/postmenopausal)	Breast cancer ascertainment (type of BCa)	Uterine cancer ascertainment	Number of Uterine / endometrial cancer in the analysis	Risk factors explored in the study
Cortesi et al. [25]	Italy	Prospective, 7512	1,704/5,808	Medical records (Invasive BCa)	Medical records	Endometrial cancer($n = 1{,}288$)	Age at BCa diagnosis, Adjuvant hormone therapy (Tamoxifen use and duration of use)
Andersson et al. [30]	Denmark	Prospective, 31,818	NA	Medical records (NA)	Medical records	Uterine cancer ($n = 183$)	Age at BCa diagnosis, Adjuvant hormone therapy (Tamoxifen use), chemotherapy, radiotherapy
Lee et al. [27]	Taiwan	Prospective, 53,783	NA	Medical records (NA)	Medical records	Uterine cancer ($n = 84$)	Age at BCa diagnosis
Brown et al. [43]	Denmark, Finland, Norway, Sweden	Prospective, 376,825	NA	Medical records (NA)	Medical records	Uterine cancer ($n = 2{,}526$)	Age at BCa diagnosis
Schaapveld et al. [31]	Netherland	Prospective, 58,068	NA	Medical records (Invasive BCa)	Medical records	Uterine cancer ($n = 310$)	Age at BCa diagnosis, Adjuvant hormone therapy (not specified), chemotherapy, radiotherapy
Trentham-Dietz et al. [12]	USA	Prospective, 10,953	2,211/8,742	Medical records (Invasive BCa)	Medical records	Endometrial cancer ($n = 113$)	BMI, adult weight gain, adult height, smoking status, alcohol intake, family history of breast cancer, age at menarche, age at fist birth, parity, oral contraceptive use, HRT use, menopausal status, age at menopause
Mellemkjaer et al. [28]	Australia, Canada, Denmark, Finland, Iceland, Norway, Scotland, Singapore, Slovenia, Sweden, Spain	Prospective 525,527	204,149/ 321,378	Medical records (NA)	Medical records	Uterine cancer ($n = 2{,}694$)	Menopausal status (Age at BCa diagnosis)
Raymond et al. [21]	USA	Prospective 332,014	NA	Medical records (NA)	Medical records	Uterine cancer (n = 2,397)	Age at BCa diagnosis
Soerjomataram et al. [33]	Netherlands	Prospective, 9919	2950/ 6969	Medical records (Invasive BCa)	Medical records	Uterine cancer ($n = 40$)	Menopausal status at BCa diagnosis

Table 2 (continued)

Author (year)	Country	Study design (N)	Menopausal status at BCa diagnosis (N pre- or peri/postmenopausal)	Breast cancer ascertainment (type of BCa)	Uterine cancer ascertainment	Number of Uterine / endometrial cancer in the analysis	Risk factors explored in the study
Swerdlow et al. [32]	England, Wales, Scotland	Case/Control study, 813 BCA survivors with ECA/ 1,067 BCA survivors without)	NA	Medical records (Invasive BCa)	Medical records	Endometrial cancer ($n=813$)	Body weight, smoking status, parity, HRT use, duration of HRT use, Adjuvant hormone therapy (Tamoxifen use and duration of use, other adjuvant hormone therapy)
Tanaka et al. [37]	Japan	Prospective, 2786	NA	Medical records (Invasive BCa)	Medical records	Uterine cancer ($n=4$)	Age at BCa diagnosis, adjuvant hormone therapy (Tamoxifen use), chemotherapy
Bergman et al. [35]	Netherland	Case/Control study, 299 BCA survivors with ECA/ 860 BCA survivors without)	Cases: 32/266 Controls: 91/753 (peri- and postmenopausal combined)	Medical records (NA)	Medical records	Endometrial cancer ($n=299$)	Adjuvant hormone therapy (Tamoxifen use and duration of use)
Mignotte et al. [36]	France	Case /Control study, 135 BCA survivors with ECA/ 487 BCA survivors without)	Cases: 49/86a Controls: 171/316a	Medical records, (NA)	Medical records	Endometrial cancer ($n=135$)	Adjuvant hormone therapy (Tamoxifen use and duration of use), chemotherapy, radiotherapy

[a]Menopausal status not defined

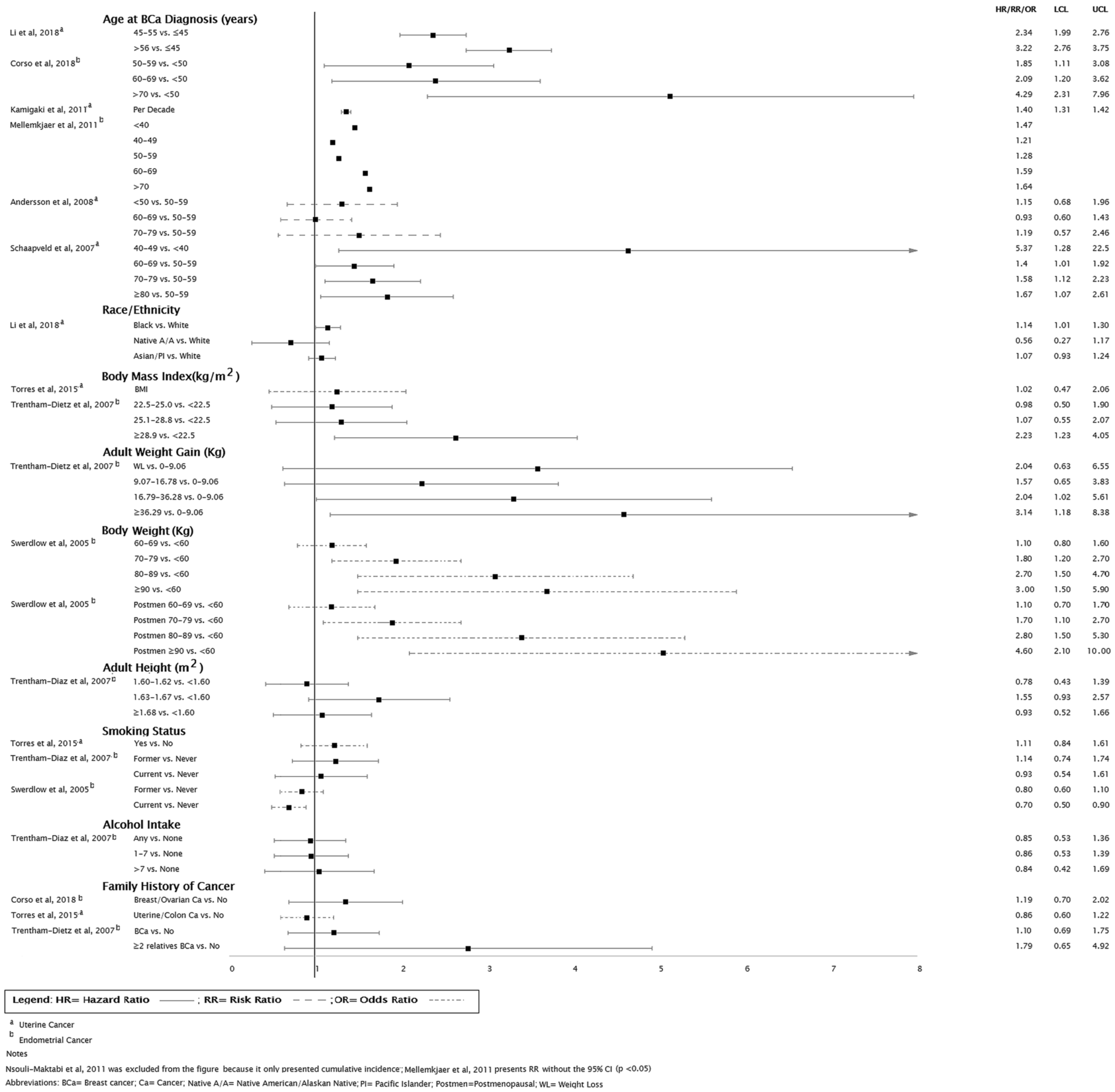

Fig. 2 Associations of selected characteristics with uterine/endometrial cancer risk among breast cancer survivors

well as among survivors with greater weight gain prior to breast cancer diagnosis (≥ 36.29 vs. 0–9.06 kg) (Fig. 2) [12]. There were no significant findings reported for adult height [12], smoking status [12, 19, 32], alcohol intake [12], and family history of cancers (breast, uterine, and/or ovarian) [6, 12, 19].

The six studies which investigated the association of menopausal status with risk of uterine cancer among breast cancer survivors had conflicting results (Figs. 3 and 5) [6, 9, 12, 19, 28, 33]. While two studies reported a stronger positive association of uterine cancers risk among pre-/perimenopausal breast cancer survivors as compared to postmenopausal breast cancer survivors [9, 33], the other 4 studies found that the association was stronger in postmenopausal as compared to premenopausal survivors [19, 28]. Although the overall association of HRT with uterine/endometrial cancer risk among breast cancer survivors was not significant, one study found a positive association with use of HRT for more than 3 months as compared to non-use [32] and another study found positive associations with estrogen only and patch estrogen HRT as compared to non-use [19]. None of the studies found significant associations for age at

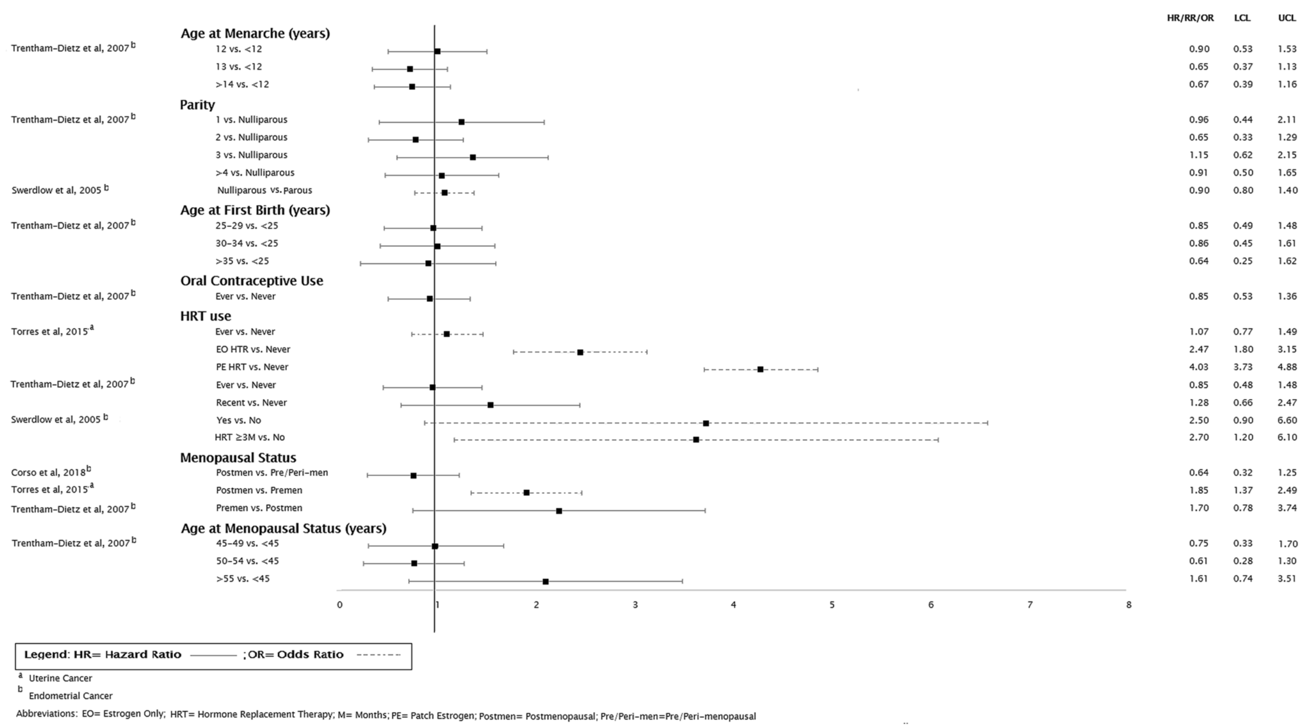

Fig. 3 Associations of reproductive factors with uterine/endometrial cancer risk among breast cancer survivors

menarche, parity, age at first birth, oral contraceptives, and age at menopausal status.

Eight out of thirteen studies reported a significant positive association between adjuvant hormone therapy and the risk of uterine or endometrial cancer among breast cancer survivors (Figs. 4 and 5) [25, 29–32, 34-36]. Nine out of thirteen studies which investigated adjuvant hormone therapy specifically explored the association with Tamoxifen use and uterine/endometrial cancer risk in breast cancer survivors and of these, six studies found a positive association [25, 30, 32, 34–36]. The positive association was most pronounced among current Tamoxifen users [35] and long-term users (> 2 years or > 5 years) [19, 32, 35–37] as compared to non-users. Furthermore, the study conducted by Chlebowski et al. reported a significantly reduced risk of endometrial cancer in women who were treated with aromatase inhibitors (AI) only for breast cancer as compared to those who used only Tamoxifen (HR 0.52, 95% CI 0.31–0.87) [38]. Selected studies also reported a significantly increased risk of uterine cancer among breast cancer survivors in women with selected comorbidities including gall bladder disease, thyroid disease, and hypertension [19] as well as for pelvic/abdominal radiotherapy as compared to no radiotherapy [32, 36]. Finally, of the 4 studies that considered histopathological characteristics of breast cancer in association with the risk of subsequent uterine/endometrial cancer, 3 were null, whereas one reported that ER/PR status was significantly associated with the risk of uterine cancer (Fig. 4) [9].

Discussion

The evidence on potential risk factors associated with uterine cancer in breast cancer survivors remains limited. We systematically reviewed 27 published studies that reported on associations between at least one potential risk factor and uterine or endometrial cancer in breast cancer survivors. Differences in study designs, including characteristics of the study population, inclusion of heterogeneous subtypes and measured/unmeasured confounders make direct comparison of the results across the studies difficult. Below, we discuss some of those methodological concerns which could have resulted in inconsistent and biased associations in the selected studies.

Insufficient number of uterine cancer cases

Even though the sample size across studies was large (> 700 participants), the small number of uterine/endometrial cancer cases among breast cancer survivors available for multivariable and stratified analysis make the results for majority of the selected studies questionable due to the lack of power to detect significant associations [39–41]. For example, the

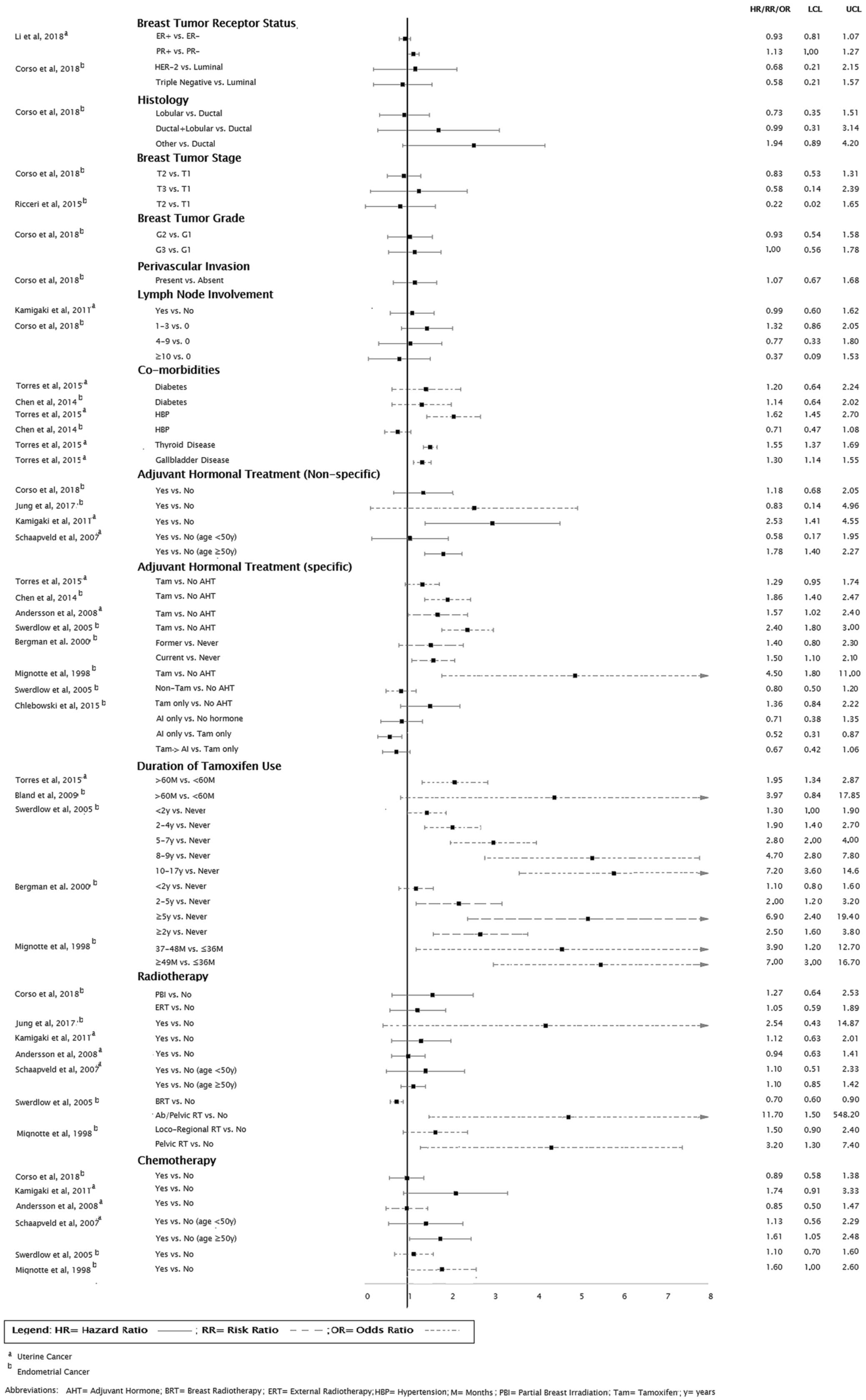

Fig. 4 Associations of clinical factors with uterine/endometrial cancer risk among breast cancer survivors

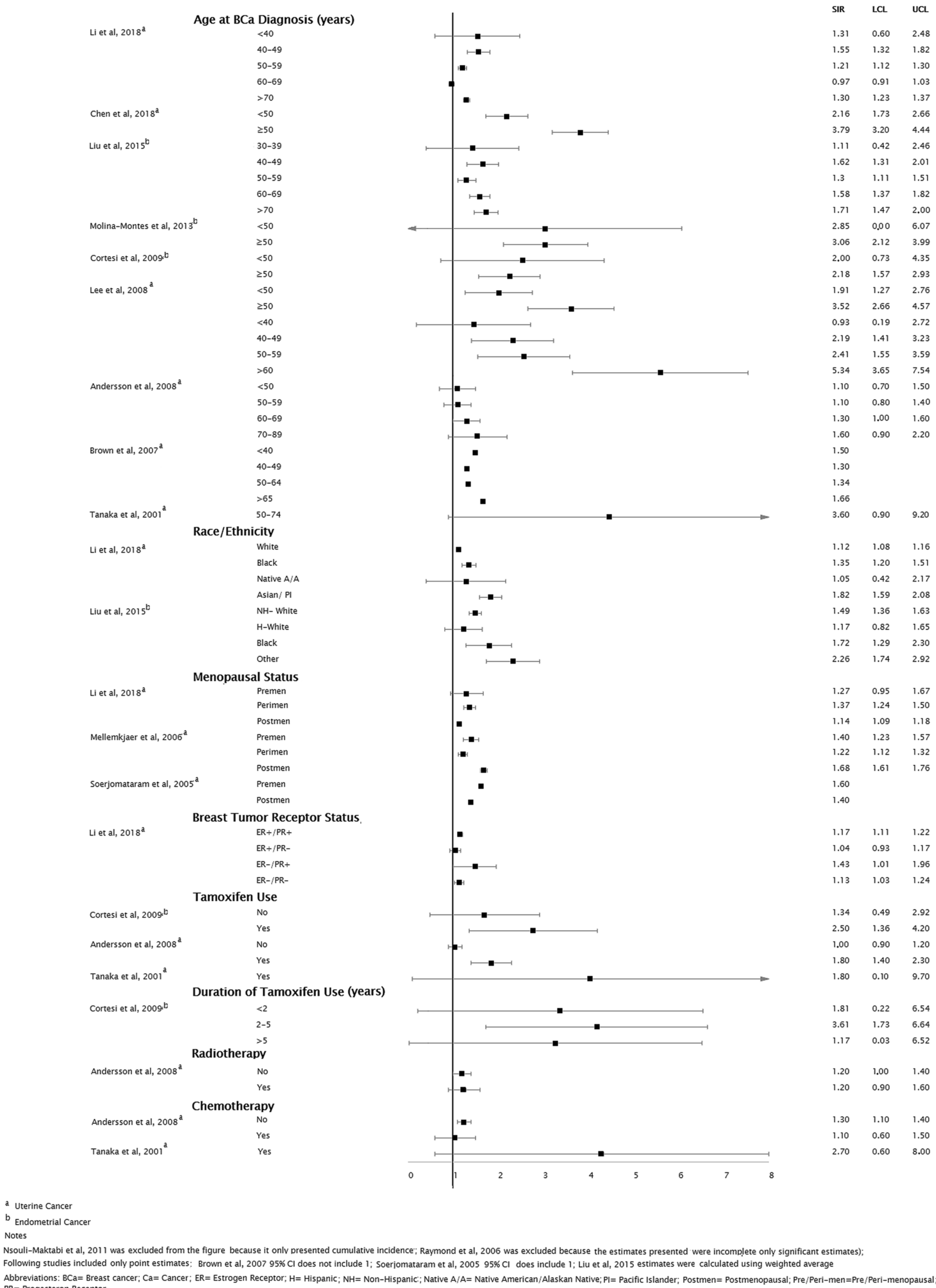

Fig. 5 Standardized Incidence Ratio for the associations of selected characteristics with uterine/endometrial cancer risk among breast cancer survivors

risk estimates of multiple potential risk factors evaluated in Corso et al., Jung et al., and Bland et al. were based on extremely small number of uterine/endometrial cancer cases ($n = 98$, 6, and 59, respectively), potentially explaining non-significant findings and wide confidence intervals.

Ascertainment of uterine/endometrial cancer cases

Most of the selected studies utilized medical records to ascertain cancer cases, either directly through tertiary medical centers/hospitals [4, 16] or through cancer registries [3, 5–7, 9, 12, 20, 21, 23–25, 27-33, 35-38, 42, 43]. Cancer registries cover a wider geographical area and multiple medical centers/hospitals; thus, it is more generalizable as compared to direct ascertainment of cases from one or few tertiary medical centers/hospitals. However, differences in completeness, coverage, timeliness and quality of cancer registries may impact generalizability and validity of findings across studies. As an example, Taiwan Cancer Registry covers 97.6% cancer patients [24, 44]. Another study conducted by Corso et al. included 33 high-quality Italian cancer registries from Cancer Incidence in Five Continents [6], and these registries only cover 70% of the Italian population [45–47]. Nevertheless, most of these cancer registries have linked multiple sources of information (death certificates, pathology reports, medical records etc.) for case ascertainment and confirmation. Although, using a single source such as insurance records may lead to case ascertainment bias or under-ascertainment, Taiwan national health insurance database utilized by Chen et al. [23] was reported to be consistent with Taiwan national cancer registry (91.5% sensitivity for all cancers) [48]. Finally, findings from the study conducted by Torres et al. which utilized self-reported cancer diagnosis must be interpreted with caution, because the authors did not verify the self-reported diagnosis for accuracy [19]. Even though previous studies consistently reported high accuracy of self-reported breast cancer diagnosis (sensitivity 85–98%) [49, 50], the accuracy of self-reported endometrial cancer was found to be much lower (sensitivity 71–15%) [49, 50].

Another potential methodological concern could be lack of consideration of uterine cancer subtypes (uterine corpus or endometrial). However, due to many shared risk factors between uterine cancer [51] and endometrial cancer [52], and majority of the uterine cancers being endometrial cancer [8], differences in the risk estimates across subtypes are expected to be negligible.

Finally, across studies there was no standard cutoff for the time between breast cancer and uterine/endometrial cancer diagnoses to distinguish co-occurring uterine/endometrial cancer (due to potentially shared risk factors with breast cancer) from subsequent primary uterine/endometrial cancers with no etiologic relatedness to breast cancer (duration cutoff for exclusion: mean = 3.9 months, range: 2–12 month). Given the wide range of cutoff values utilized by selected studies, it is possible that some of these studies included cases of uterine/endometrial cancer that concurrently developed with breast cancer and were diagnosed shortly after breast cancer diagnosis. Therefore, standardization of cutoff values to separate co-occurring uterine/endometrial from subsequent primary uterine/endometrial cancers in future studies is warranted.

Important confounders and other potential risk factors

Previous studies suggest that obesity [11–13], genetic predisposition (as defined by family history of breast, uterine and ovarian cancer) [14, 15, 53], hormonal and reproductive factors (age at menarche, oral contraceptive use, parity, menopausal status, postmenopausal hormone use) [9, 16] may influence the risk of uterine/endometrial cancer among breast cancer survivors. However, only a limited number of studies examined multiple potential risk factors in the same study population and several studies failed to account for possible important confounders or provide clear descriptions of the risk factors and their definitions included in the multivariable models. As an example, two studies [4, 9] did not discuss the individual factors included in the multivariable models and/or their data collection approaches and variable definitions. Another study [19] had conflicting description of statistical methods with reference to multivariate adjustment and model selection in the methods section and justification of not using multivariate adjustment due to the small sample size in the results, thus making it unclear whether the presented risk estimates represent crude or adjusted results. While only few selected studies clearly specified each of the variable in their multivariable models, these studies did not include important potential confounders such as BMI [30, 36], menopausal status [30, 36], HRT [30, 36], and adjuvant therapy [12]. Furthermore, some studies did not specify whether they utilized univariable or multivariable analysis [5, 23, 25, 27, 28, 37, 43]. Taking into account these data and statistical methodology gaps, proper adjustment for potential confounding factors in future studies is warranted.

Next, inconsistent findings on the associations of lifestyle and reproductive factors could be partially explained by the use of different risk factor definitions across studies. As an example, only a limited number of studies reported risk estimates for menopausal status, and of these, only one study clearly defined menopausal status by considering either self-reported natural menopause or combination of age, history of hysterectomy, and smoking status [12]. The remaining studies either did not define menopausal status or utilized age as a proxy for its definition. Use of age as the only factor for definition of menopausal status without consideration of other important factors such as surgical or medication-induced menopause and/or smoking history could result in significant misclassification [54, 55]. For example, one study [12] included women with hysterectomy in the postmenopausal category which may also potentially lead to misclassification as women with hysterectomy without oophorectomy would still be considered premenopausal. Finally, due to physiological and lifestyle differences between pre- and postmenopausal women [56–58] it is crucial to estimate risk of uterine cancer among breast cancer survivors by well-defined categories of menopausal status.

Comparison of risk factors for primary uterine/endometrial cancer in the general population vs. in breast cancer survivors

Comparison of the risk factors between primary uterine/endometrial cancer in the general population and in breast cancer survivors could inform cancer prevention needs that are specific for breast cancer survivors. Comparison of the risk factors for primary uterine/endometrial cancer in the general population vs. in breast cancer survivors is presented in Table 3.

Across several epidemiological studies, age was consistently reported to be positively associated with endometrial/uterine cancer in the general population [52, 59] as well as among breast cancer survivors [6, 9, 23, 27, 29, 31, 42, 43, 60] suggesting that age is a nonspecific risk factor of uterine/endometrial cancer in breast cancer survivors. Several studies reported lower incidence of uterine/endometrial or endometrial cancer in African American as compared to white women [61–64]. In contrast, African American breast cancer survivors seem to have an increased risk of uterine/endometrial cancer as compared to white breast cancer survivors [9, 20]. Thus, race could be a specific risk factor of uterine/endometrial cancer in breast cancer survivors.

Obesity and adult weight gain are well-recognized risk factors for uterine/endometrial cancer in the general population [52, 59, 65–70]. Based on existing data, obesity and adult weight gain seem to be positively associated with endometrial cancer risk in breast cancer survivors. The evidence suggests that obesity, and adult weight gain are not specific risk factors of uterine/endometrial cancer among breast cancer survivors. However, only a limited number of studies explored the associations of obesity and adult weight gain with the risk of uterine/endometrial cancer in breast cancer survivors; thus, these results should to be interpreted with caution. Association between smoking histories was inversely associated with uterine/endometrial cancer in the general population [59, 69, 71]. In contrast, the three studies which explored the associations between smoking history and uterine/endometrial cancer risk among breast cancer survivors reported inconsistent findings. The majority of studies also found no significant association between alcohol intake and uterine/endometrial cancer in the general population [72–74]. Although only Trentham-Dietz et al. study explored this association among breast cancer survivors, findings were consistent with reports from the general population [12]. Two reviews reported that family history of endometrial or colorectal cancer is positively associated with the risk of endometrial cancer among women in the general population [59, 75]. Family history of cancer was defined in three studies variously as family history of breast cancer, family history of breast and ovarian cancer, or family history of colorectal and uterine cancer; thus, we could not compare the results across these studies.

Several studies found associations between reproductive risk factors and the risk of uterine/endometrial cancer in the general population, including a positive association with age at first birth, HRT use, being postmenopausal and age at menopause as well as an inverse association with age at menarche, parity and oral contraceptive use (Table 3). However, among breast cancer survivors these associations were either not significant (parity), inconsistent (HRT use, menopausal status) or have limited evidence (age at first birth, oral contraceptive use, age at menopause). In addition, previous studies reported positive associations of some comorbidities such as diabetes and hypertension with uterine/endometrial cancer risk among women in the general population (Table 3), but evidence on these associations among breast cancer survivors is extremely limited.

Finally, despite the methodological concerns discussed above, the majority of epidemiological studies consistently reported positive association between tamoxifen use and uterine/endometrial cancer in breast cancer survivors. However, prior studies also demonstrate benefits of tamoxifen use in reducing the risk of contralateral breast cancer and breast cancer recurrence among breast cancer survivors [76, 77]. Therefore, further investigations are urgently needed to determine if the benefits of tamoxifen use in breast cancer survivors outweigh the risk of uterine/endometrial cancer."

Conclusion

In general, we found limited epidemiological data on associations between potential risk factors and uterine/endometrial cancer among breast cancer survivors. Based on a comprehensive review of the available literature, we found that older age, obesity and increased adult body weight appear to be shared risk factors for uterine/endometrial cancer in the general population and in breast cancer survivors. Our review also suggests that African American breast cancer survivors might be at an increased risk of uterine/endometrial cancer following breast cancer diagnosis. In addition, Tamoxifen might be a risk factor specific for uterine/endometrial cancer risk among breast cancer survivors. Future studies should

Table 3 Comparison of risk factors for primary uterine cancer/endometrial cancer in breast cancer survivors and in the general population

Risk factor	Studies with significant findings and direction of association among breast cancer survivors (+ or −)	Studies with non-significant findings among breast cancer survivors	Risk factor for uterine/endometrial cancer in the general population (Yes [+ or −], No or Limited evidence)
Age at BCa diagnosis	Li [9] (+) Corso [6] (+) Kamigaki [29] (+) Mellemkjaer [28] (+) Schaapveld [31] (+) Chen [24] (+) Liu [20] (+) Molina-Montes [5] (+) Cortesi [25] (+) Lee [27] (+) Brown [43] (+)	Andersson [30] Tanaka [37]	Plagens-Rotman [60] (+)[b] Ali [59] (+)[b]
Race (African American vs. White)	Li [9] (+) Higher risk for African American Liu [20] (+) Higher risk for African American	–	Setiawan [61] (−)Higher risk for White women
BMI	Trentham [12] (+)	Torres [19]	American Cancer Society 2019 (+) Raglan [65] (+)[b] Birmann [69] (+) Arnold [67] (+) Aune [66] (+)[b] Jenabi [70] (+)[b] Nevadunsky [68] (+) Ali [59] (+)[b]
Weight gain	Trentham [12] (+)	–	Aune [66] (+)[b]
Adult weight	Swerdlow [32] (+)	–	–
Adult height	–	Trentham [12]	Aune [66] (+)[b]
Smoking history	Swerdlow [32] (−)	Torres [19] Trentham [12]	Ali [59] (−)[b] Birmann [69] (−) Zhou [71] (−)[b]
Alcohol intake	–	Trentham [12]	Birmann [69] (−) Je 2014 (−) Zhou [73] (no association)b Fedirko [72] (no association) Friberg [74] (no association)[b]
Family history of cancer	–	Corso [6] Torres [19] Trentham [12]	American Cancer Society 2019 (+) (Relatives with endometrial or colon cancer) Win [75] (+)[b] (First degree family history of endometrial and colorectal cancer) Ali [59] (+)[b]
Early age at menarche	–	Trentham [12]	Sponholtz 2017 (+) Birmann [69] (+) Gong 2015 (−)[b] -Late age Ali 2014 (+)[b] Ali [59] (+)[b] Zucchetto 2009 (−) Late age
Late age at first birth	–	Trentham [12]	Sponholtz 2017 (+) Ali 2014 (+)b Zucchetto 2009 (no association)

Table 3 (continued)

Risk factor	Studies with significant findings and direction of association among breast cancer survivors (+ or −)	Studies with non-significant findings among breast cancer survivors	Risk factor for uterine/endometrial cancer in the general population (Yes [+ or −], No or Limited evidence)
Parity	–	Trentham [12] Swerdlow [32]	American Cancer Society 2019 (−) Raglan [65] (−)[b] Sponholtz 2017 (−) Birmann [69] (−) Wu 2015 (−) Ali 2014 (−)[b] Ali [59] (+)[b] Nulliparity
HRT	Torres [19] (+) Swerdlow [32] (+)	Torres [19] Trentham [12] Swerdlow [32]	Sjogren 2016 (+) Ali [59] (+)[b] Birmann [69] (+) Zucchetto 2009 (no association)
Menopausal status (Peri- or Postmenopausal)	Torres [19] (+) Mellemkjaer [28] (+) Li [9] (+)	Corso [6] Trentham [12] Soerijomataram 2005	Gao 2016 (+) Ali [59] (+)[b] Zucchetto 2009 (no association)
Late age at menopause	–	Trentham [12]	Ali 2014 (+)[b] Zucchetto 2009 (+)
Oral contraceptive use	–	Trentham [12]	American Cancer Society 2019 (−) Collaborative Group on Epidemiological Studies on Endometrial Cancer 2015 (−)[b] Ali 2014 (−)[b] Zucchetto 2009 (−)
Comorbidities	Torres [19] (+) (Hypertension)[a]	Torres [19] (Diabetes) Chen [23] (Diabetes) Chen [23] (Hypertension)	American Cancer Society 2019 (+) (Diabetes) Friberg 2007 (+) (Diabetes)b Aune 2017 (+) (Hypertension)b Sun 2015 (+) (Hypertension) Ali [59][b] (inconclusive for diabetes and hypertension)

[a]In addition, Torres et al. reported positive associations for Thyroid and Gallbladder disease

[b]A narrative/ systematic review or a meta-analysis

focus on a wider range of factors that may influence the risk of uterine/endometrial cancer among breast cancer survivors while addressing methodological concerns discussed in the current review. Specifically, robust methods for ascertainment of uterine/endometrial cancer diagnosis should be implemented along with standardized definition of co-occurring vs. primary cancers and important study variables (for example, menopausal status) should be clearly defined to reduce misclassification. Further, detailed description of the methods across individual studies including statistical modeling and adjustment for confounders would facilitate future meta-analysis. Finally, studies with sufficiently larger numbers of uterine/endometrial cancer cases are needed to be able to conduct more powerful investigations and to perform proper stratified analyses, at least by women's menopausal status.

Acknowledgements None.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Informed consent Not applicable.

Research involving human participants and/or animals Not applicable.

References

1. American Cancer Society ACS (2018) Cancer Facts & Figures, Atlanta
2. Surveillance E, and End Results Program (2018) Female Breast Cancer—Cancer Stat Facts. https://seer.cancer.gov/statfacts/html/breast.html.
3. Ricceri F, Fasanelli F, Giraudo MT, Sieri S, Tumino R, Mattiello A, Vagliano L, Masala G, Quiros JR, Travier N, Sanchez MJ, Larranaga N, Chirlaque MD, Ardanaz E, Tjonneland A, Olsen

A, Overvad K, Chang-Claude J, Kaaks R, Boeing H, Clavel-Chapelon F, Kvaskoff M, Dossus L, Trichopoulou A, Benetou V, Adarakis G, Bueno-de-Mesquita HB, Peeters PH, Sund M, Andersson A, Borgquist S, Butt S, Weiderpass E, Skeie G, Khaw KT, Travis RC, Rinaldi S, Romieu I, Gunter M, Kadi M, Riboli E, Vineis P, Sacerdote C (2015) Risk of second primary malignancies in women with breast cancer: results from the European prospective investigation into cancer and nutrition (EPIC). Int J Cancer 137(4):940–948
4. Jung HK, Park S, Kim NW, Lee JE, Kim Z, Han SW, Hur SM, Kim SY, Lim CW, Lee MH, Lee J (2017) Development of second primary cancer in Korean breast cancer survivors. Ann Surg Treat Res 93(6):287–292
5. Molina-Montes E, Pollan M, Payer T, Molina E, Davila-Arias C, Sanchez MJ (2013) Risk of second primary cancer among women with breast cancer: a population-based study in Granada (Spain). Gynecol Oncol 130(2):340–345
6. Corso G, Veronesi P, Santomauro GI, Maisonneuve P, Morigi C, Peruzzotti G, Intra M, Sacchini V, Galimberti V (2018) Multiple primary non-breast tumors in breast cancer survivors. J Cancer Res Clin Oncol 144(5):979–986
7. Nsouli-Maktabi HH, Henson DE, Younes N, Young HA, Cleary SD (2011) Second primary breast, endometrial, and ovarian cancers in Black and White breast cancer survivors over a 35-year time span: effect of age. Breast Cancer Res Treat 129(3):963–969
8. Casey MJ, Crotzer D (2019) Cancer, Endometrial. In: StatPearls (ed) StatPearls Publishing LLC, Treasure Island (FL)
9. Li Z, Wu Q, Song J, Zhang Y, Zhu S, Sun S (2018) Risk of second primary female genital malignancies in women with breast cancer: a SEER analysis. Horm Cancer 9(3):197–204
10. Ngo C, Brugier C, Plancher C, de la Rochefordiere A, Alran S, Feron JG, Malhaire C, Scholl S, Sastre X, Rouzier R, Fourchotte V (2014) Clinico-pathology and prognosis of endometrial cancer in patients previously treated for breast cancer, with or without tamoxifen: a comparative study in 363 patients. Eur J Surg Oncol 40(10):1237–1244
11. Molina-Montes E, Requena M, Sanchez-Cantalejo E, Fernandez MF, Arroyo-Morales M, Espin J, Arrebola JP, Sanchez MJ (2015) Risk of second cancers cancer after a first primary breast cancer: a systematic review and meta-analysis. Gynecol Oncol 136(1):158–171
12. Trentham-Dietz A, Newcomb PA, Nichols HB, Hampton JM (2007) Breast cancer risk factors and second primary malignancies among women with breast cancer. Breast Cancer Res Treat 105(2):195–207
13. Druesne-Pecollo N, Touvier M, Barrandon E, Chan DS, Norat T, Zelek L, Hercberg S, Latino-Martel P (2012) Excess body weight and second primary cancer risk after breast cancer: a systematic review and meta-analysis of prospective studies. Breast Cancer Res Treat 135(3):647–654
14. Segev Y, Rosen B, Lubinski J, Gronwald J, Lynch HT, Moller P, Kim-Sing C, Ghadirian P, Karlan B, Eng C, Gilchrist D, Neuhausen SL, Eisen A, Friedman E, Euhus D, Ping S, Narod SA (2015) Risk factors for endometrial cancer among women with a BRCA1 or BRCA2 mutation: a case control study. Fam Cancer 14(3):383–391
15. Lee YC, Milne RL, Lheureux S, Friedlander M, McLachlan SA, Martin KL, Bernardini MQ, Smith C, Picken S, Nesci S, Hopper JL, Phillips KA (2017) Risk of uterine cancer for BRCA1 and BRCA2 mutation carriers. Eur J Cancer 84:114–120
16. Bland AE, Calingaert B, Secord AA, Lee PS, Valea FA, Berchuck A, Soper JT, Havrilesky L (2009) Relationship between tamoxifen use and high risk endometrial cancer histologic types. Gynecol Oncol 112(1):150–154
17. Dong C, Chen L (2014) Second malignancies after breast cancer: the impact of adjuvant therapy. Mol Clin Oncol 2(3):331–336
18. Kirova YM, De Rycke Y, Gambotti L, Pierga JY, Asselain B, Fourquet A, Institut Curie Breast Cancer Study G (2008) Second malignancies after breast cancer: the impact of different treatment modalities. Br J Cancer 98(5):870–874
19. Torres D, Myers JA, Eshraghi LW, Riley EC, Soliman PT, Milam MR (2015) Risk factors for the development of uterine cancer in breast cancer survivors: an army of women study. Ann Surg Oncol 22(6):1974–1979
20. Liu J, Jiang W, Mao K, An Y, Su F, Kim BY, Liu Q, Jacobs LK (2015) Elevated risks of subsequent endometrial cancer development among breast cancer survivors with different hormone receptor status: a SEER analysis. Breast Cancer Res Treat 150(2):439–445
21. Raymond JS, Hogue CJR (2006) Multiple primary tumours in women following breast cancer, 1973–2000. Br J Cancer 94(11):1745–1750. https://doi.org/10.1038/sj.bjc.6603172
22. Partners SRaLRSbE (2019) DistillerSR forest plot generator. Evidence Partners, Canada
23. Chen JY, Kuo SJ, Liaw YP, Avital I, Stojadinovic A, Man YG, Mannion C, Wang J, Chou MC, Tsai HD, Chen ST, Hsiao YH (2014) Endometrial cancer incidence in breast cancer patients correlating with age and duration of tamoxifen use: a population based study. J Cancer 5(2):151–155
24. Chen MC, Lee KD, Lu CH, Wang TY, Huang SH, Chen CY (2018) The bidirectional association among female hormone-related cancers: breast, ovary, and uterine corpus. Cancer Med 7(6):2299–2306
25. Cortesi L, De Matteis E, Rashid I, Cirilli C, Proietto M, Rivasi F, Federico M (2009) Distribution of second primary malignancies suggests a bidirectional effect between breast and endometrial cancer: a population-based study. Int J Gynecol Cancer 19(8):1358–1363
26. Liu X, Zhao F, Hu L, Sun Y (2015) Value of detection of serum human epididymis secretory protein 4 and carbohydrate antigen 125 in diagnosis of early endometrial cancer of different pathological subtypes. Onco Targets Ther 8:1239–1243
27. Lee KD, Chen SC, Chan CH, Lu CH, Chen CC, Lin JT, Chen MF, Huang SH, Yeh CM, Chen MC (2008) Increased risk for second primary malignancies in women with breast cancer diagnosed at young age: a population-based study in Taiwan. Cancer Epidemiol Biomark Prev 17(10):2647–2655
28. Mellemkjaer L, Friis S, Olsen JH, Scelo G, Hemminki K, Tracey E, Andersen A, Brewster DH, Pukkala E, McBride ML, Kliewer EV, Tonita JM, Kee-Seng C, Pompe-Kirn V, Martos C, Jonasson JG, Boffetta P, Brennan P (2006) Risk of second cancer among women with breast cancer. Int J Cancer 118(9):2285–2292
29. Kamigaki Y, Kawakami K (2011) Risk of second cancer after initial treatment of breast cancer: an osaka cancer registry database study. Oncol Lett 2(5):963–973
30. Andersson M, Jensen MB, Engholm G, Henrik Storm H (2008) Risk of second primary cancer among patients with early operable breast cancer registered or randomised in Danish Breast Cancer Cooperative Group (DBCG) protocols of the 77, 82 and 89 programmes during 1977–2001. Acta Oncol 47(4):755–764
31. Schaapveld M, Visser O, Louwman MJ, de Vries EG, Willemse PH, Otter R, van der Graaf WT, Coebergh JW, van Leeuwen FE (2008) Risk of new primary nonbreast cancers after breast cancer treatment: a Dutch population—based study. J Clin Oncol 26(8):1239–1246
32. Swerdlow AJ, Jones ME (2005) Tamoxifen treatment for breast cancer and risk of endometrial cancer: a case–control study. J Natl Cancer Inst 97(5):375–384

33. Soerjomataram I, Louwman WJ, de Vries E, Lemmens VEPP, Klokman WJ, Coebergh JWW (2005) Primary malignancy after primary female breast cancer in the South of the Netherlands, 1972–2001. Breast Cancer Res Treat 93(1):91–95
34. Chen JY, Kuo SJ, Liaw YP, Avital I, Stojadinovic A, Man YG, Mannion C, Wang J, Chou MC, Tsai HD, Chen ST (2014) Endometrial cancer incidence in breast cancer patients correlating with age and duration of tamoxifen use: a population based study. J Cancer 5(2):e151
35. Bergman L, Beelen ML, Gallee MP, Hollema H, Benraadt J, van Leeuwen FE (2000) Risk and prognosis of endometrial cancer after tamoxifen for breast cancer comprehensive cancer centres' ALERT Group Assessment of liver and endometrial cancer risk following tamoxifen. Lancet 356(9233):881–887
36. Mignotte H, Lasset C, Bonadona V, Lesur A, Luporsi E, Rodier JF, Cutuli B, Lasry S, Mauriac L, Granon C, Kerr C, Giard S, Hill C, de Lafontan B, de Gislain C, D'Anjou J, Fondrinier E, Lefeuvre C, Parache RM, Chauvin F (1998) Iatrogenic risks of endometrial carcinoma after treatment for breast cancer in a large French case-control study federation nationale des centres de lutte contre le cancer (FNCLCC). Int J Cancer 76(3):325–330
37. Tanaka H, Tsukuma H, Koyama H, Kinoshita Y, Kinoshita N, Oshima A (2001) Second primary cancers following breast cancer in the Japanese female population. Jpn J Cancer Res 92(1):1–8
38. Chlebowski RT, Schottinger JE, Shi J, Chung J, Haque R (2015) Aromatase inhibitors, tamoxifen, and endometrial cancer in breast cancer survivors. Cancer 121(13):2147–2155
39. Hajian-Tilaki K (2011) Sample size estimation in epidemiologic studies. Caspian J Intern Med 2(4):289–298
40. Peduzzi P, Concato J, Kemper E, Holford TR, Feinstein AR (1996) A simulation study of the number of events per variable in logistic regression analysis. J Clin Epidemiol 49(12):1373–1379
41. Noordzij M, Tripepi G, Dekker FW, Zoccali C, Tanck MW, Jager KJ (2010) Sample size calculations: basic principles and common pitfalls. Nephrol Dial Transpl 25(5):1388–1393
42. Mellemkjaer L, Christensen J, Frederiksen K, Pukkala E, Weiderpass E, Bray F, Friis S, Andersson M, Olsen JH (2011) Risk of primary non-breast cancer after female breast cancer by age at diagnosis. Cancer Epidemiol Biomark Prev 20(8):1784–1792
43. Brown LM, Chen BE, Pfeiffer RM, Schairer C, Hall P, Storm H, Pukkala E et al (2007) Risk of second non-hematological malignancies among 376,825 breast cancer survivors. Breast Cancer Res Treat 106(3):439–451
44. Lee J, Park S, Kim S, Kim J, Ryu J, Park HS, Kim SI, Park BW (2015) Characteristics and survival of breast cancer patients with multiple synchronous or metachronous primary cancers. Yonsei Med J 56(5):1213–1220
45. Bray F, Ferlay J (2019) Chapter 5: data comparability and quality. World Health Organization, IARC. https://ci5.iarc.fr/CI5-XI/Pages/Chapter5.aspx.
46. Association ICR (2017) Map of cancer registries. https://www.registri-tumori.it/cms/pagine/mappa-dei-registri-di-popolazione.
47. Antoni S, Soerjomataram I, Møller B, Bray F, Ferlay J (2016) An assessment of GLOBOCAN methods for deriving national estimates of cancer incidence. Bull World Health Organ 94(3):174
48. Kao WH, Hong JH, See LC, Yu HP, Hsu JT, Chou IJ, Chou WC, Chiou MJ, Wang CC, Kuo CF (2018) Validity of cancer diagnosis in the National Health Insurance database compared with the linked National Cancer Registry in Taiwan. Pharmacoepidemiol Drug Saf 27(10):1060–1066
49. Parikh-Patel A, Allen M, Wright WE (2003) Validation of self-reported cancers in the California Teachers Study. Am J Epidemiol 157(6):539–545
50. Navarro C, Chirlaque MD, Tormo MJ, Pérez-Flores D, Rodríguez-Barranco M, Sánchez-Villegas A, Agudo A, Pera G, Amiano P, Dorronsoro M, Larrañaga N, Quirós JR, Ardanaz E, Barricarte A, Martínez C, Sánchez MJ, Berenguer A, González CA (2006) Validity of self reported diagnoses of cancer in a major Spanish prospective cohort study. J Epidemiol Community Health 60(7):593–599. https://doi.org/10.1136/jech.2005.039131
51. Oncology ASoC (2012) Uterine cancer—risk factors and prevention. https://www.cancer.net/cancer-types/uterine-cancer/risk-factors-and-prevention.
52. Society AC (2019) Endometrial cancer risk factors.
53. Marcheselli R, Marcheselli L, Cortesi L, Bari A, Cirilli C, Pozzi S, Ferri P, Napolitano M, Federico M, Sacchi S (2015) Risk of second primary malignancy in breast cancer survivors: a nested population-based case–control study. J Breast Cancer 18(4):378–385
54. Phipps AI, Ichikawa L, Bowles EJ, Carney PA, Kerlikowske K, Miglioretti DL, Buist DS (2010) Defining menopausal status in epidemiologic studies: a comparison of multiple approaches and their effects on breast cancer rates. Maturitas 67(1):60–66
55. Trentham-Dietz A, Sprague BL, Hampton JM, Miglioretti DL, Nelson HD, Titus LJ, Egan KM, Remington PL, Newcomb PA (2014) Modification of breast cancer risk according to age and menopausal status: a combined analysis of five population-based case-control studies. Breast Cancer Res Treat 145(1):165–175
56. Eden JA (2013) Menopausal status, adipose tissue, and breast cancer risk: impact of estrogen replacement therapy. Horm Mol Biol Clin Investig 14(2):57–63
57. Bonotto M, Puglisi F (2015) Early breast cancer in pre-menopausal women and endocrine treatment: which factors impact on decision-making process? Recent Prog Med 106(8):364–369
58. Sr J (2014) Menopausal hormone therapy and breast cancer. J Steroid Biochem Mol Biol 142:52–61
59. Ali AT (2013) Risk factors for endometrial cancer. Ceska Gynekol 78(5):448–459
60. Plagens-Rotman K, Żak E, Pięta B (2016) Odds ratio analysis in women with endometrial cancer. Prz Menopauzalny 15(1):12–19. https://doi.org/10.5114/pm.2016.58767
61. Setiawan VW, Pike MC, Kolonel LN, Nomura AM, Goodman MT, Henderson BE (2006) Racial/ethnic differences in endometrial cancer risk: the Multiethnic Cohort Study. Am J Epidemiol 165(3):262–270
62. Allard JE, Maxwell GL (2009) Race disparities between black and white women in the incidence, treatment, and prognosis of endometrial cancer. Cancer Control 16(1):53–56
63. Cote ML, Alhajj T, Ruterbusch JJ, Bernstein L, Brinton LA, Blot WJ, Chen C, Gass M, Gaussoin S, Henderson B, Lee E, Horn-Ross PL, Kolonel LN, Kaunitz A, Liang X, Nicholson WK, Park AB, Petruzella S, Rebbeck TR, Setiawan VW, Signorello LB, Simon MS, Weiss NS, Wentzensen N, Yang HP, Zeleniuch-Jacquotte A, Olson SH (2015) Risk factors for endometrial cancer in black and white women: a pooled analysis from the epidemiology of Endometrial Cancer Consortium (E2C2). Cancer Causes Control 26(2):287–296
64. Babatunde OA, Adams SA, Eberth JM, Wirth MD, Choi SK, Hebert JR (2016) Racial disparities in endometrial cancer mortality-to-incidence ratios among blacks and whites in South Carolina. Cancer Causes Control 27(4):503–511
65. Raglan O, Kalliala I, Markozannes G, Cividini S, Gunter MJ, Nautiyal J, Gabra H, Paraskevaidis E, Martin-Hirsch P, Tsilidis KK, Kyrgiou M (2018) Risk factors for endometrial cancer: an umbrella review of the literature. Int J Cancer 145(7):1719–1730
66. Aune D, Navarro Rosenblatt DA, Chan DS, Vingeliene S, Abar L, Vieira AR, Greenwood DC, Bandera EV, Norat T (2015) Anthropometric factors and endometrial cancer risk: a systematic review and dose-response meta-analysis of prospective studies. Ann Oncol 26(8):1635–1648
67. Arnold M, Pandeya N, Byrnes G, Renehan PAG, Stevens GA, Ezzati PM, Ferlay J, Miranda JJ, Romieu I, Dikshit R, Forman D, Soerjomataram I (2015) Global burden of cancer attributable to

high body-mass index in 2012: a population-based study. Lancet Oncol 16(1):36–46
68. Nevadunsky NS, Van Arsdale A, Strickler HD, Moadel A, Kaur G, Levitt J, Girda E, Goldfinger M, Goldberg GL, Einstein MH (2014) Obesity and age at diagnosis of endometrial cancer. Obstet Gynecol 124(2 Pt 1):300–306
69. Birmann BM, Barnard ME, Bertrand KA, Bao Y, Crous-Bou M, Wolpin BM, De Vivo I, Tworoger SS (2016) Nurses' health study contributions on the epidemiology of less common cancers: endometrial, ovarian, pancreatic, and hematologic. Am J Public Health 106(9):1608–1615. https://doi.org/10.2105/ajph.2016.303337
70. Jenabi E, Poorolajal J (2015) The effect of body mass index on endometrial cancer: a meta-analysis. Public Health 129(7):872–880
71. Zhou B, Yang L, Sun Q, Cong R, Gu H, Tang N, Zhu H, Wang B (2008) Cigarette smoking and the risk of endometrial cancer: a meta-analysis. Am J Med 121(6):501–508.e503
72. Fedirko V, Jenab M, Rinaldi S, Biessy C, Allen NE, Dossus L, Onland-Moret NC, Schutze M, Tjonneland A, Hansen L, Overvad K, Clavel-Chapelon F, Chabbert-Buffet N, Kaaks R, Lukanova A, Bergmann MM, Boeing H, Trichopoulou A, Oustoglou E, Barbitsioti A, Saieva C, Tagliabue G, Galasso R, Tumino R, Sacerdote C, Peeters PH, Bueno-de-Mesquita HB, Weiderpass E, Gram IT, Sanchez S, Duell EJ, Molina-Montes E, Arriola L, Chirlaque MD, Ardanaz E, Manjer J, Lundin E, Idahl A, Khaw KT, Romaguera-Bosch D, Wark PA, Norat T, Romieu I (2013) Alcohol drinking and endometrial cancer risk in the European Prospective Investigation into Cancer and Nutrition (EPIC) study. Ann Epidemiol 23(2):93–98
73. Zhou Q, Guo P, Li H, Chen XD (2017) Does alcohol consumption modify the risk of endometrial cancer? A dose-response meta-analysis of prospective studies. Arch Gynecol Obstet 295(2):467–479
74. Friberg E, Orsini N, Mantzoros CS, Wolk A (2010) Alcohol intake and endometrial cancer risk: a meta-analysis of prospective studies. Br J Cancer 103(1):127–131
75. Win AK, Reece JC, Ryan S (2015) Family history and risk of endometrial cancer: a systematic review and meta-analysis. Obstet Gynecol 125(1):89–98
76. Hackshaw A, Roughton M, Forsyth S, Monson K, Reczko K, Sainsbury R, Baum M (2011) Long-term benefits of 5 years of tamoxifen: 10-year follow-up of a large randomized trial in women at least 50 years of age with early breast cancer. J Clin Oncol 29(13):1657–1663
77. Gray RG, Rea D, Handley K, Bowden SJ, Perry P, Earl HM, Poole CJ, Bates T, Chetiyawardana S, Dewar JA, Fernando IN, Grieve R, Nicoll J, Rayter Z, Robinson A, Salman A, Yarnold J, Bathers S, Marshall A, Lee M, Group obotaC (2013) aTTom: long-term effects of continuing adjuvant tamoxifen to 10 years versus stopping at 5 years in 6,953 women with early breast cancer. J Clin Oncol 31:5–5